

WEEK 7 REPORT - STUDENT PERFORMANCE

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INTRODUCTION:

In Week 7, the Student Performance in Exams dataset was loaded into Power BI and prepared for analysis using Power Query. The student details and subject scores were checked, cleaned and transformed. Calculated columns and DAX measures were then created so that student performance could be understood easily.

DATA IMPORTING:

The dataset was imported into Power BI and opened in the Power Query Editor. The number of rows and columns was checked to understand the structure of the data. The data types, missing values, errors and duplicate records were also reviewed. This helped to find any problems in the dataset before the cleaning step.

StudentsPerformance.csv							
File Origin		Delimiter		Data Type Detection			
1252: Western European (Windows)		Comma		Based on first 200 rows			
gender	race/ethnicity	parental level of education	lunch	test preparation course	math score	reading score	writing score
female	group B	bachelor's degree	standard	none	72	72	74
female	group C	some college	standard	completed	69	90	88
female	group B	master's degree	standard	none	90	95	93
male	group A	associate's degree	free/reduced	none	47	57	44
male	group C	some college	standard	none	76	78	75
female	group B	associate's degree	standard	none	71	83	78
female	group B	some college	standard	completed	88	95	92
male	group B	some college	free/reduced	none	40	43	39
male	group D	high school	free/reduced	completed	64	64	67
female	group B	high school	free/reduced	none	38	60	50
male	group C	associate's degree	standard	none	58	54	52
male	group D	associate's degree	standard	none	40	52	43
female	group B	high school	standard	none	65	81	73
male	group A	some college	standard	completed	78	72	70
female	group A	master's degree	standard	none	50	53	58
female	group C	some high school	standard	none	69	75	78
male	group C	high school	standard	none	88	89	86
Extract Table Using Examples							
				Load	Transform Data	Cancel	

DATA CLEANING:

The dataset was cleaned by removing duplicate records and checking for invalid values. Data types were corrected wherever needed, for example setting the score columns as whole numbers. The categorical values were standardized so that the text entries were clear and consistent.

A ^B _C gender	A ^B _C race/ethnicity	A ^B _C parental level of education	A ^B _C lunch
Valid 100%	Valid 100%	Valid 100%	Valid
Error 0%	Error 0%	Error 0%	Error
Empty 0%	Empty 0%	Empty 0%	Empty
Female	Group B	Bachelor'S Degree	Standard
Female	Group C	Some College	Standard
Female	Group B	Master'S Degree	Standard
Male	Group A	Associate'S Degree	Free/Reduced
Male	Group C	Some College	Standard
Female	Group B	Associate'S Degree	Standard
Female	Group B	Some College	Standard
Male	Group B	Some College	Free/Reduced
Male	Group D	High School	Free/Reduced
Female	Group B	High School	Free/Reduced
Male	Group C	Associate'S Degree	Standard

CALCULATED COLUMNS:

New columns were created to measure each student's overall performance. The Total Score column adds the Math, Reading and Writing marks, and the Average Score column gives the mean of the three subjects.

Custom Column

Add a column that is computed from the other columns.

New column name

Total Marks

Custom column formula ⓘ

= [math score]+[reading score]+[writing score]

Available columns

gender

race/ethnicity

parental level of education

lunch

test preparation course

math score

reading score

writing score

<< Insert

Custom Column

Add a column that is computed from the other columns.

New column name

Average Score

Custom column formula ⓘ

= [Total Marks]/3

Available columns

race/ethnicity
parental level of education
lunch
test preparation course
math score
reading score
writing score
Total Marks

<< Insert

PERFORMANCE GRADE AND PASS/FAIL RESULT:

A Performance Grade column was also created using conditional logic to classify students into performance levels. The Performance Grade classifies students into different levels based on their scores, which makes it easy to see who performed well and who needs improvement. A Pass/Fail column was also added to show whether a student passed or failed based on the selected pass criteria.

Add Conditional Column

Add a conditional column that is computed from the other columns or values.

New column name

Grade

	Column Name	Operator	Value ⓘ		Output ⓘ
If	Average Score	is greater than or...	80	Then	A
Else If	Average Score	is greater than or...	60	Then	B
Else If	Average Score	is greater than or...	40	Then	C

Add Clause

Else ⓘ

Fail

OK

Cancel

Add Conditional Column

Add a conditional column that is computed from the other columns or values.

New column name
Result

	Column Name	Operator	Value		Output	
If	Average	is greater than or...	40	Then	Pass	...

Add Clause

Else
Fail

OK Cancel

SUBJECT DATA TRANSFORMATION:

The Math, Reading and Writing score columns were converted from a wide format to a long format using the Unpivot Columns option in Power Query. The new columns were renamed Subject and Score, which makes it easier to compare performance across subjects. A duplicate table named Student Details was also created, containing only the student information without the subject marks. Keeping the student details separate from the score data makes the dataset cleaner and easier to manage in the data model.

123 Student ID	A ^B C Subject	123 Score
Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
1	Math Score	72
1	Reading Score	72
1	Writing Score	74
2	Math Score	69
2	Reading Score	90
2	Writing Score	88
3	Math Score	90
3	Reading Score	95
3	Writing Score	93
4	Math Score	47
4	Reading Score	57
4	Writing Score	44

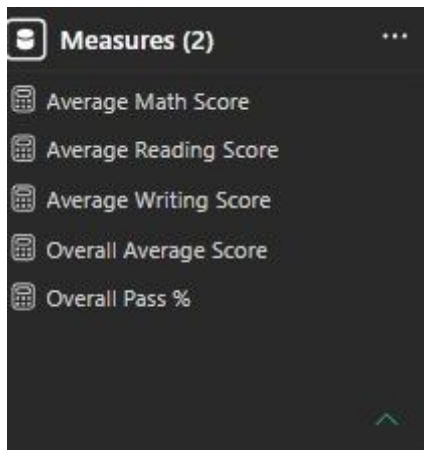
DAX MEASURES AND DATA MODEL:

The transformed tables were loaded into the Power BI data model. Several DAX measures were created for the analysis: Average Math Score, Average Reading Score, Average Writing Score, Overall Average Score and Overall Pass %. These measures give quick summary values and can be reused in visuals and reports later.

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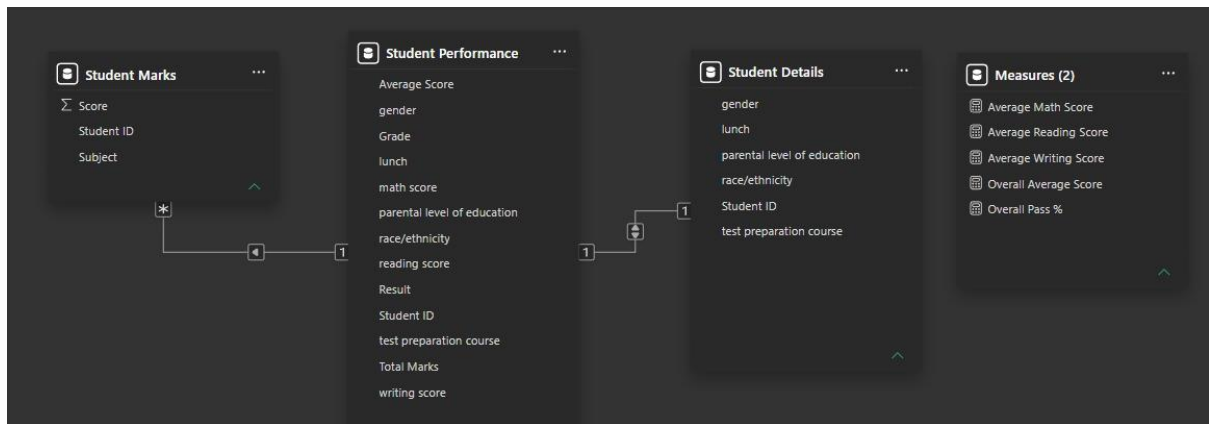
Average Math Score = AVERAGE('Student Performance'[math score])
Average Reading Score = AVERAGE('Student Performance'[reading score])
Average Writing Score = AVERAGE('Student Performance'[writing score])
Average Writing Score = AVERAGE('Student Performance'[writing score])
Overall Average Score = AVERAGE('Student Marks'[Score])
Overall Pass % = DIVIDE(CALCULATE(COUNT('Student Performance'[Result]),'Student Performance'[Result]="Pass"),DISTINCTCOUNT('Student Performance'[Student ID]),0)

```



FINAL OUTPUT:

The final output is a cleaned and transformed Student Performance dataset. It includes calculated fields such as Total Score, Average Score, Performance Grade and Pass/Fail classification. The subject scores were unpivoted into a Subject and Score format, and the basic DAX measures were created to support further analysis.



CONCLUSION:

In Week 7, the Student Performance dataset was successfully cleaned and transformed using Power Query and Power BI. Calculated columns, Performance Grades and Pass/Fail results were created to understand how students performed. The subject data was unpivoted for easier comparison, and DAX measures were added to calculate the key performance values. The final dataset is now ready for visualization and deeper analysis.